Q1 - 24 June - Shift 1

0.056 kg of Nitrogen is enclosed in a vessel at a temperature of 127°C. The amount of heat required to double the speed of its molecules is ______ k cal.

(Take $R = 2$ cal $mole^{-1}K^{-1}$)

Space for your notes:

Q2 - 25 June - Shift 1

The relation between root mean square speed (v_{rms}) and most probable speed (v_p) for the molar mass M of oxygen gas molecule at the temperature of 300 K will be :-

(A) $v_{rms} = \sqrt{\frac{2}{3}} v_p$

(B) $v_{rms} = \sqrt{\frac{3}{2}} v_p$

(C) $v_{rms} = v_p$

(D) $v_{rms} = \sqrt{\frac{1}{3}} v_p$

Space for your notes:

Q3 - 25 June - Shift 2

The ratio of specific heats $\left(\frac{C_P}{C_V}\right)$ in terms of degree of freedom (f) is given by:

(A) $\left(1+\frac{f}{3}\right)$

(B) $\left(1+\frac{2}{f}\right)$

(C) $\left(1+\frac{f}{2}\right)$

(D) $\left(1+\frac{1}{f}\right)$

Space for your notes:

Q4 - 25 June - Shift 2

When a gas filled in a closed vessel is heated by raising the temperature by 1°C, its pressure increase by 0.4%. The initial temperature of the gas is______ K.

Space for your notes:

Q5 - 26 June - Shift 2

A flask contains argon and oxygen in the ratio of 3:2 in mass and the mixture is kept at 27°C. The ratio of their average kinetic energy per molecule respectively will be :

(A) 3 : 2
(B) 9 : 4
(C) 2 : 3
(D) 1 : 1

Space for your notes:

Q6 - 27 June - Shift 1

A mixture of hydrogen and oxygen has volume 2000 cm^3, temperature 300 K, pressure 100 kPa and mass 0.76 g The ratio of number of moles of hydrogen to number of moles of oxygen in the mixture will be :

(A) $\frac{1}{3}$
(B) $\frac{3}{1}$
(C) $\frac{1}{16}$
(D) $\frac{16}{1}$

Space for your notes:

Q7 - 27 June - Shift 2

According to kinetic theory of gases,

A. The motion of the gas molecules freezes at $0^{\circ}C$

B. The mean free path of gas molecules decreases if the density of molecules is increased.

C. The mean free path of gas molecules increases if temperature is increased keeping pressure constant.

D. Average kinetic energy per molecule per degree of freedom is $\frac{3}{2}k_BT$ (for monoatomic gases)

Choose the most appropriate answer from the options given below:

(A) A and C only (B) B and C only

(C) A and B only (D) C and D only

Space for your notes:

Q8 - 28 June - Shift 1

Given below are two statement :

Statement – I : What μ amount of an ideal gas undergoes adiabatic change from state (P_1, V_1, T_1) to state (P_2, V_2, T_2), the work done is $W = \frac{1R(T_2 - T_1)}{1-\gamma}$, where $\gamma = \frac{C_P}{C_V}$ and $R =$ universal gas constant,

Statement — II: In the above case. when work is done on the gas. the temperature of the gas would rise.

Choose the correct answer from the options given below:

(A) Both statement—I and statement-II are true.

(B) Both statement—I and statement-II are false.

(C) Statement-I is true but statement-II is false.

(D) Statement-I is false but statement-II is true.

Space for your notes:

Q9 - 28 June - Shift 2

What will be the effect on the root mean square velocity of oxygen molecules if the temperature is doubled and oxygen molecule dissociates into atomic oxygen?

(A) The velocity of atomic oxygen remains same

(B) The velocity of atomic oxygen doubles

(C) The velocity of atomic oxygen becomes half

(D) The velocity of atomic oxygen becomes four times

Space for your notes:

Q10 - 29 June - Shift 2

A vessel contains 16g of hydrogen and 128 g of oxygen at standard temperature and pressure. The volume of the vessel in cm^3 is :

(A) 72×10^5 (B) 32×10^5

(C) 27×10^4 (D) 54×10^4

Space for your notes:

Answer Key

Q1 (10)	**Q2** (B)	**Q3** (B)	**Q4** (250)
Q5 (D)	**Q6** (B)	**Q7** (B)	**Q8** (A)
Q9 (B)	**Q10** (C)		

Q1 (10)

0.056 kg $N_2 = 56$ gm of $N_2 = 2$ mole of N_2

$T_1 = 400$ K, $v \alpha \sqrt{T}$ so $T_2 = 4T_1 = 1600K$

$$Q = \frac{f}{2} nR\Delta T$$

$f = 5$

$Q = 12$ k cal

Q2 (B)

$$v_{rms} = \sqrt{\frac{3RT}{M}} \text{ and } v_{mp} = \sqrt{\frac{2RT}{M}}$$

Thus $v_{rms} = \sqrt{\frac{3}{2}} v_{mp}$

Q3 (B)

Molar heat capacity at constant volume $C_V = \frac{fR}{2}$

where f is degree of freedom.

Molar heat capacity at constant pressure can be written as $C_P = R + C_V = R + \frac{fR}{2} = \left(1 + \frac{f}{2}\right)R$

So $\frac{C_P}{C_v} = 1 + \frac{2}{f}$

Q4 (250)

$$pV = nRT$$

$$\Delta P . V = nR\Delta T$$

$$\Rightarrow \frac{\Delta P}{P} = \frac{\Delta T}{T} = \frac{0.4}{100}$$

$$\Rightarrow T = \frac{100 \times 1}{0.4} = 250K$$

Q5 (D)

Average K.E./molecule $= \frac{f}{2} kT$

So, $\frac{K_{Ar}}{K_{O_2}} = \frac{\frac{3}{2}kT}{\frac{5}{2}kT} = \frac{3}{5}$

Q6 (B)

$$PV = nRT$$

$$n = \frac{100\times10^{3}\times2000\times10^{-6}}{\frac{25}{3}\times300}$$

$$n = 80\times10^{-3}$$

$$n_1 + n_2 = 0.08$$

$$n_1\times2 + n_2\times32 = 0.76$$

$$(0.08 - n_2)2 + n_2(32) = 0.76$$

$$n_2 = 0.02$$

$$n_1 = 0.06$$

$$\frac{n_1}{n_2} = \frac{3}{1}$$

Q7 (B)

$$\lambda = \frac{kT}{\sqrt{2}\pi d^2 P}$$

Q8 (A)

$$W_{adiabatic} = \frac{NR(T_f - T_i)}{1-\gamma} \rightarrow \text{statment 1}$$

$$Q = W + \Delta U$$

$$0 = W + \Delta U$$

$$\Delta U = -W$$

If work is done on the gas, i.e. work is negative

$\therefore \Delta U$ is positive.

$\therefore$ Temperature will increase.

Q9 (B)

$$V_{rms} = \sqrt{\frac{3RT}{M}}$$

$$T \rightarrow 2T$$

$$M \rightarrow \frac{M}{2}$$

$$V_{rms} \propto \sqrt{\frac{T}{M}}$$

$$\Rightarrow (V_{rms})_{atomic} = (V_{rms})_{molecular} \times \sqrt{\frac{2}{1/2}} = 2(V_{rms})_{molecular}$$

Q10 (C)

No of moles of H_2 = 8 moles

No of moles of O_2 = 4 moles

Total moles = 12 moles

At STP 1 mole occupy = $22.4\ell = 22.4 \times 10^3\ cm^3$

12 moles will occupy = $12 \times 22.4 \times 10^3\ cm^3$

$\approx 26.8 \times 10^4\ cm^3$